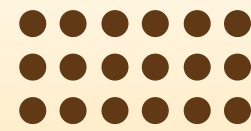




Crystal Mind



Automatic Chicken Coop Door



We keep your chicks safe

Coop GUARD



**MT1940 – Fundamental of Engineering
Design and Workshop Practice**



30/05/2025

01

What if there was a simple,
automatic way to ensure my
chickens were always safe
and on time?



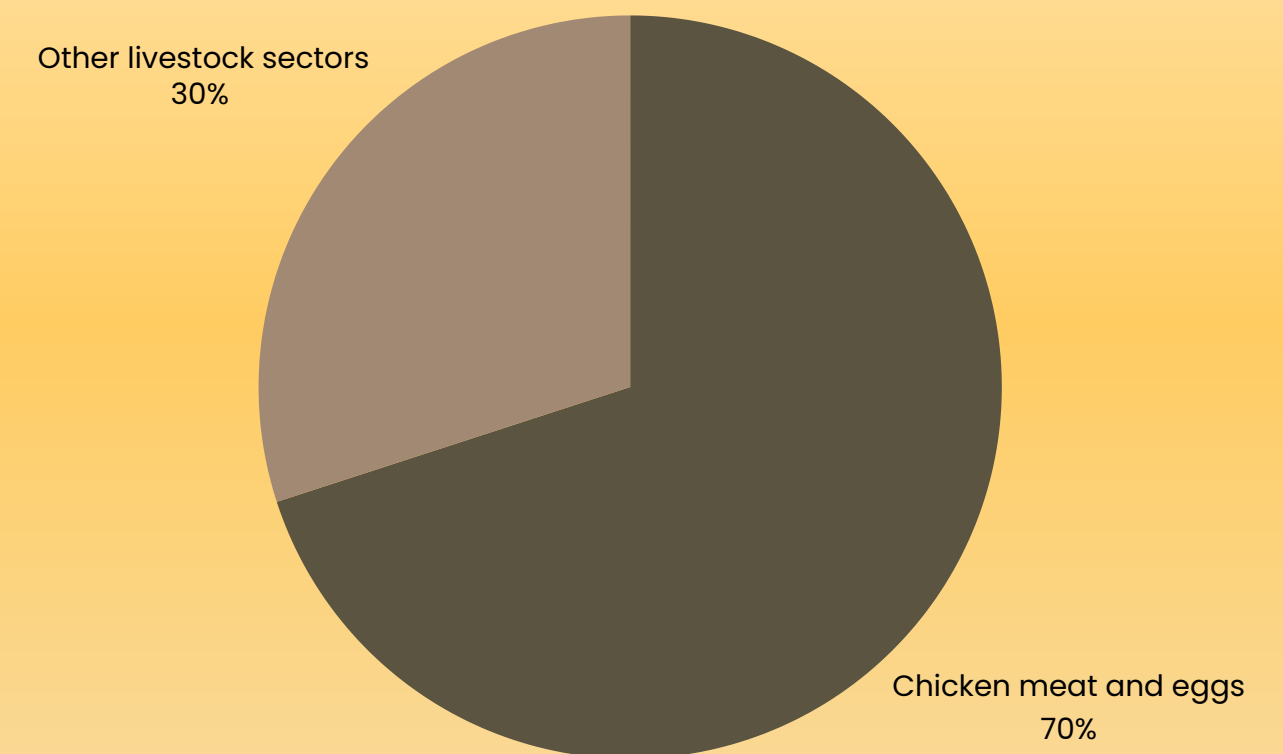
A Day in the Life of a Rural Farmer

- Morning chores, job, forgets to open the coop
- Evening forgotten – mongoose attack
- Impact: Lost food, income, and nutrition

Why Poultry Matters in Sri Lanka?

📈 “70% of the livestock sector’s value comes from chicken meat and eggs”

🥚 “15% of Sri Lanka’s total egg production comes from backyard poultry”



PROBLEM STATEMENT

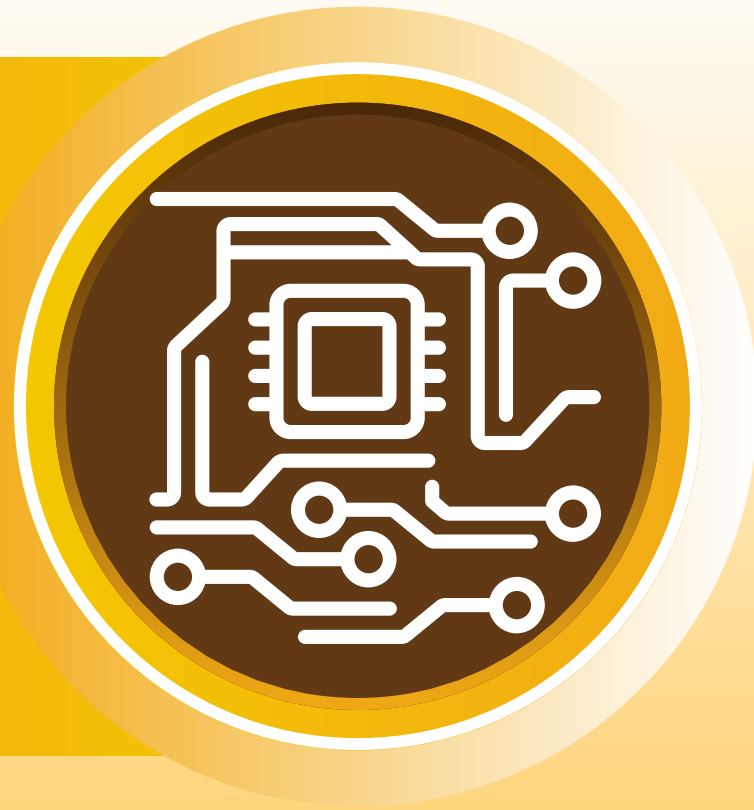
Harsh weather and tight schedules make it difficult for farmers to manage daily coop access, often leading to forgotten locks and increased vulnerability for livestock at night. This constant need for monitoring also restricts their freedom and travel flexibility.



Design Requirements

- Affordability
- Ease of use
- Safety and reliability .
- Rural applicability
- Expandability

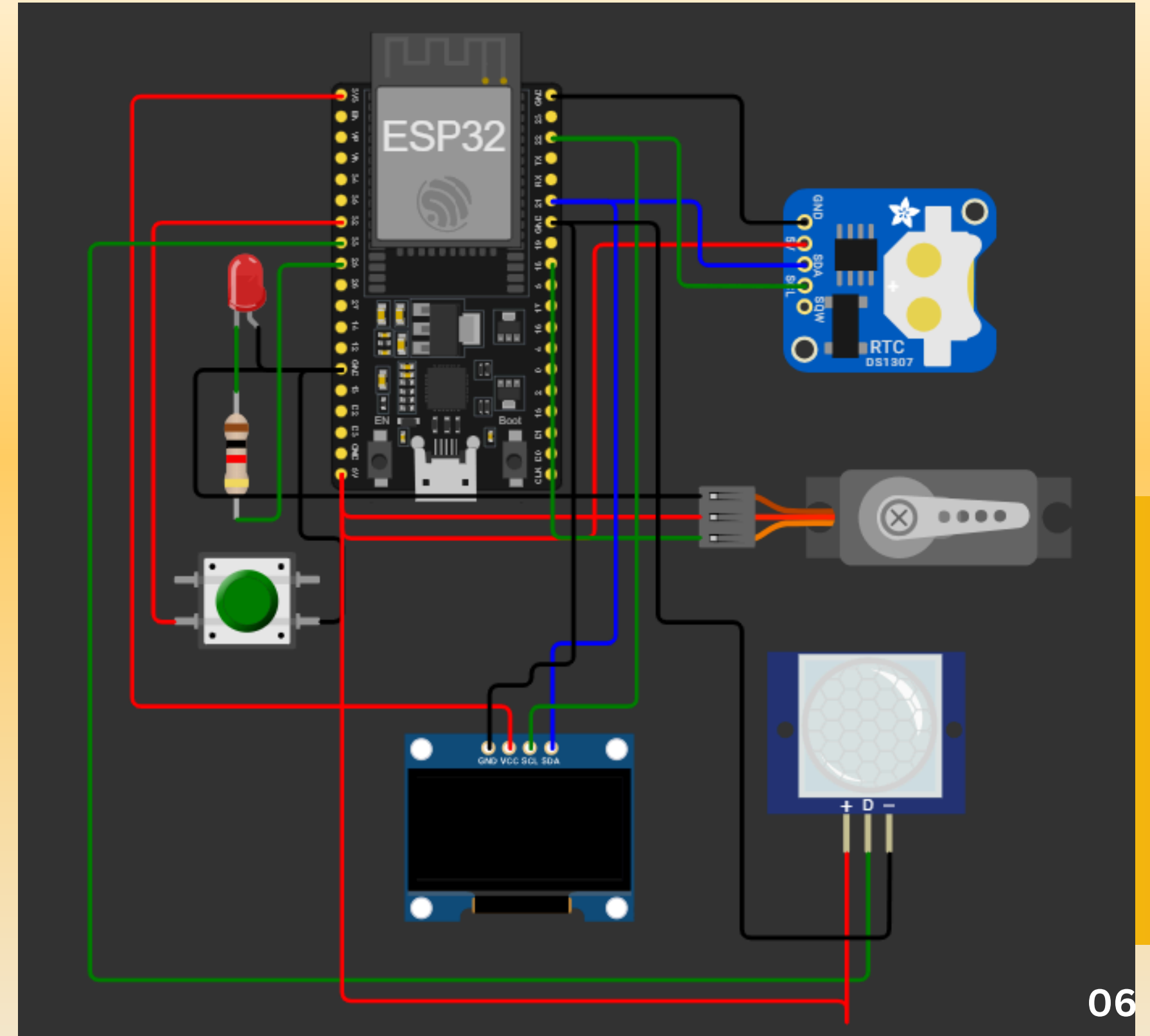
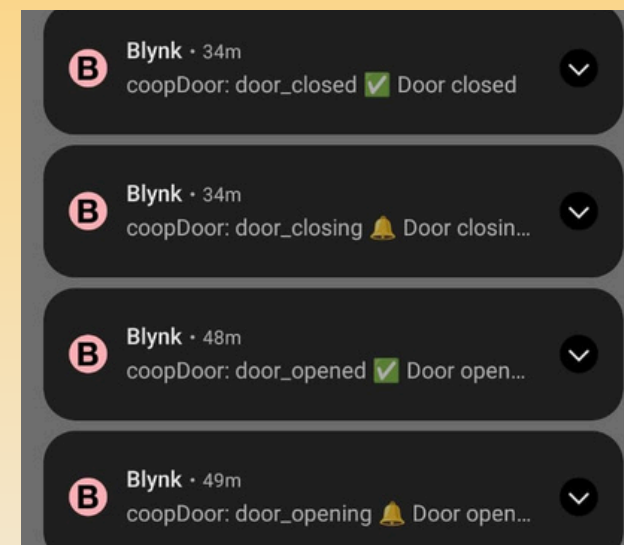




DESIGN SOLUTION

Schematic Diagram (Electronics)

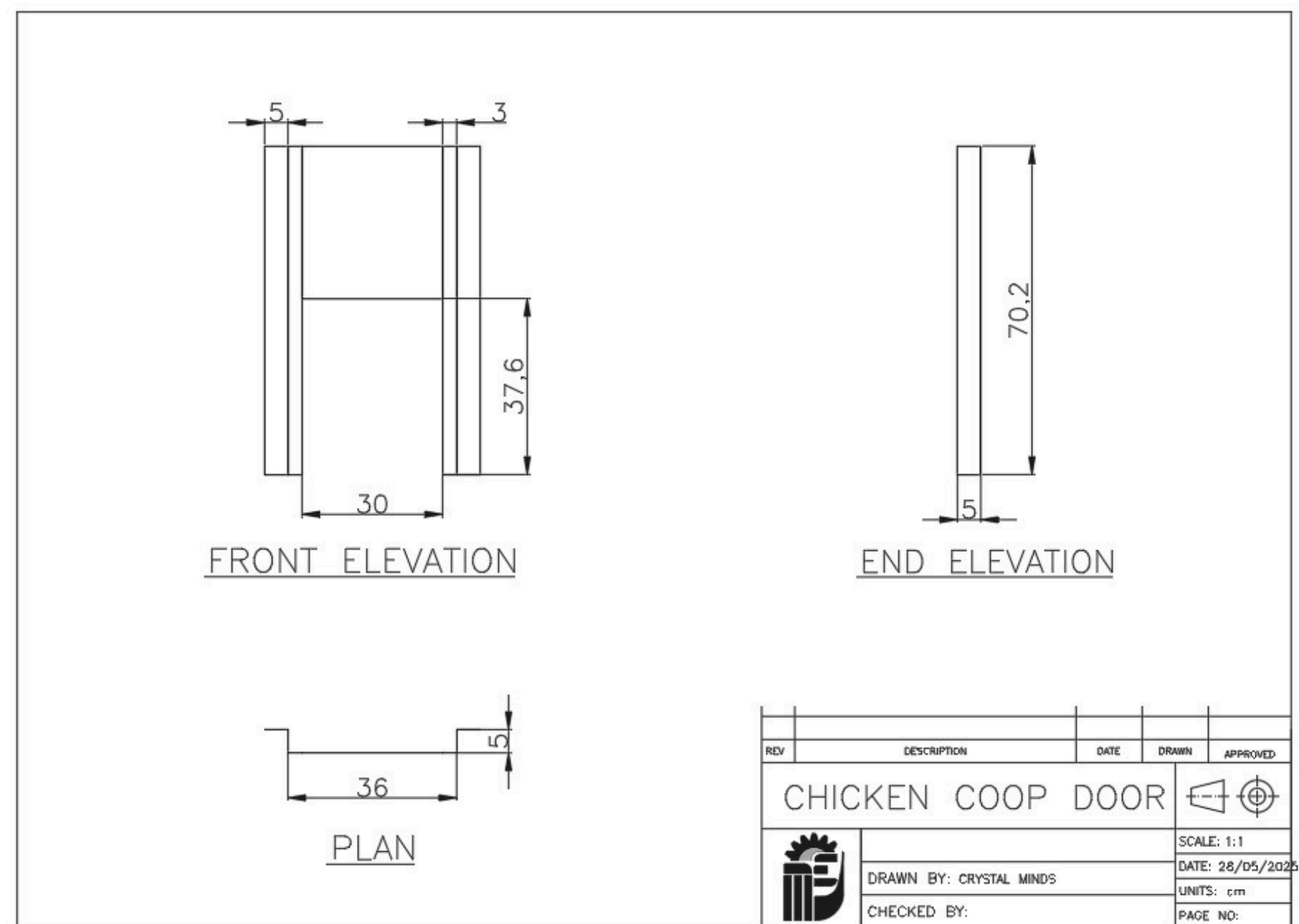
- ESP8266 microcontroller
- RTC module
- Gear motor
- LCD
- Pushbutton
- Blynk app





DESIGN SOLUTION

Mechanical Drawing



Financial Analysis Methods and Techniques



Category	Budget
Stainless steel and Al sheets	Rs. 3000
ESP8266 microcontroller	Rs. 790
RTC Module	Rs. 210
L298N motor controller	Rs. 490
Push Button	Rs. 10
OLED display	Rs. 600
Total	Rs. 5100

The Prototype in Action



Safety In Design

Key safety considerations,

- **Electrical short circuiting**
- **Motor overheating**
- **Mechanical jamming**
- **Human/animal injury**



Implementation and Deployment Strategy



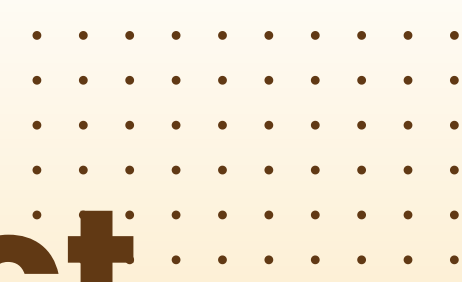
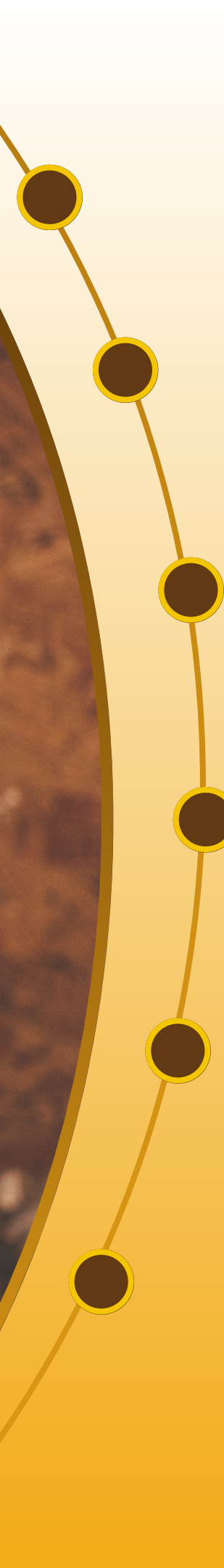
**Initial testing
in three
villages**



**Iterate and
Refine**



**Micro-Distribution
Model**



Economic Impact

- **Enhanced food security**
- **Increased farmer income**
- **National agricultural resilience**

Social & Environmental Relevance



Energy Efficiency



Improved Animal Welfare



Education & Innovation



Time & Labor Saving



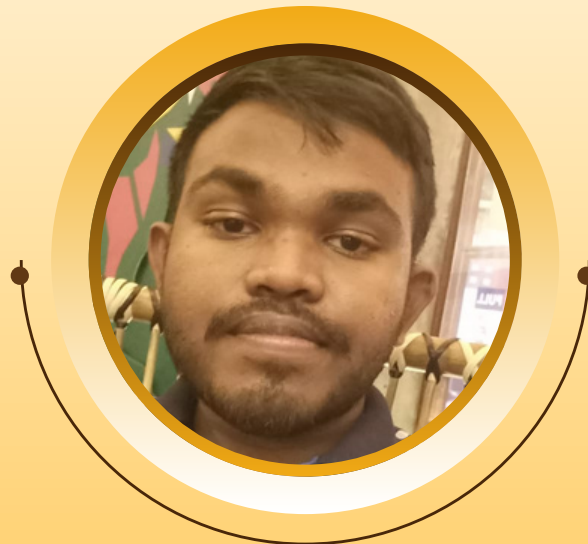
Conclusion

- Our project demonstrates how :
 - Thoughtful design
 - Empathy for stakeholders
 - Strong engineering principles can combine to deliver an elegant scalable solution to a common rural challenge.
- The Automatic Chicken Coop Door System provides:
 - Timely protection for poultry
 - Convenience for owners
 - Simplicity in engineering and operation
- It's effective in both:
 - Offline rural village
 - Connected urban areas



Crystal Minds

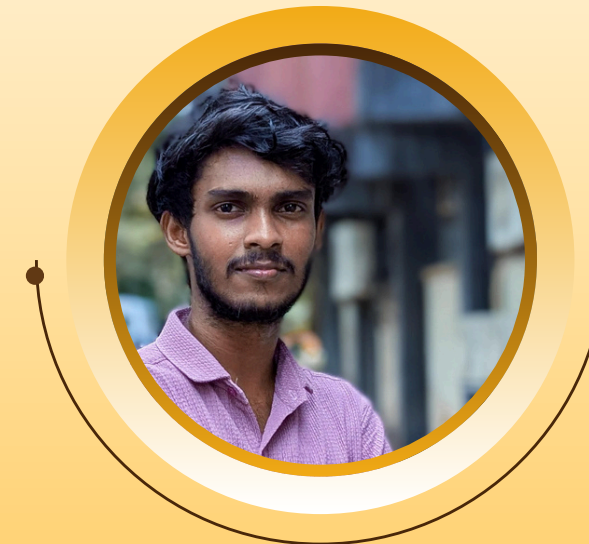
Instructed by: Mr. Themiya Jayalath
Ms. Yasara De Zoysa



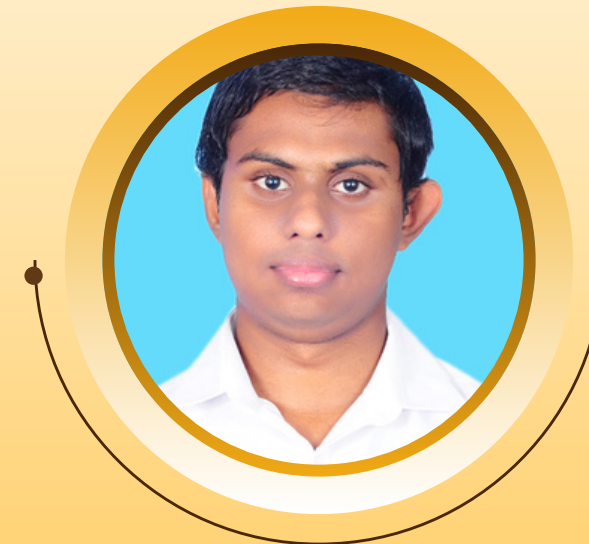
Gangul Wijerathna
Team Leader



Laksandi Denipitiya
Convener



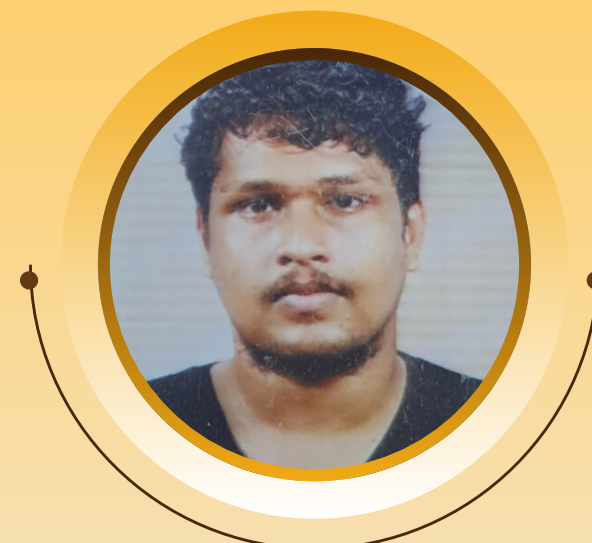
Kaween Hasaranga
Team Member



Kalana Nethsara
Team Member



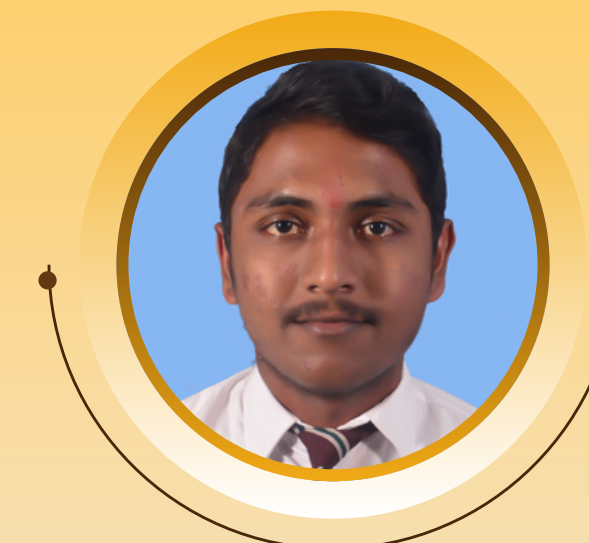
S.Subidya
Team Member



V. Kishalan
Team Member



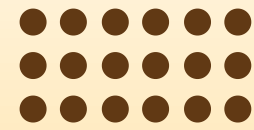
Tachini Lakshari
Team Member



A, Akihmsan
Team Member



Crystal Minds



Thank You!

FOR YOUR ATTENTION



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References

Research on avian circadian rhythms shows that chickens, like many birds, have an internal clock synchronized with the sun. This means they instinctively return to roost when it gets dark.

Reference: Gwinner, E. (1996). "Circadian and circannual rhythms in birds." *Current Opinion in Neurobiology*, 6(6), 734–738.

Poultry Behavior Studies

Studies on poultry behavior confirm that chickens develop a routine based on environmental cues, such as light levels and pre-trained habits.

Reference: Appleby, M. C., Mench, J. A., & Hughes, B. O. (2004). *Poultry Behaviour and Welfare*. CABI Publishing.

Practical Farming Guides

Many poultry farmers report that chickens reliably return to their coop at dusk without human intervention. If trained with food or artificial lighting, they can also learn to enter at a pre-set time.

Reference: Smith, P. (2011). *Storey's Guide to Raising Chickens*. Storey Publishing.